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PARTICULARS	DETAILS
Department	Artificial Intelligence & Data Science
Programme	B.Tech - Artificial Intelligence & Data Science
Course Code & Course Name	DSA0404 – Fundamentals of Data Science
Academic Year/Batch	2026-27
Faculty Name	Dr. Krishnaraj Mony
Assignment Title	
Date of Issue	31 August 2026
Date of Submission	02 September 2026
Maximum Marks	100
Course Outcomes	To understand the concept of supervised and unsupervised methods in Machine Learning
Bloom's Taxonomy Level	BL5
SDG Mapping (SDG 1-17)	SDG 9
Industry/Societal Relevance	

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1. PROBLEM UNDERSTANDING AND FORMULATION

Crop selection is influenced by the physical and environmental conditions of the farmland. Soil nutrient levels and climatic factors can vary considerably between locations, making it difficult to select an appropriate crop using a single rule. A data-driven approach can assist in identifying the crop that best matches a given set of farming conditions.

The objective of this work is to develop a Crop Selection Advisor using a CART (Classification and Regression Tree) model. The system uses six measurable conditions as inputs: Nitrogen (N), Phosphorus (P), Potassium (K), rainfall, temperature, and humidity. The corresponding crop label is treated as the target variable.

The problem is formulated as a multi-class classification task, since the model must select one crop category from the available crop classes based on the supplied input conditions.

The agricultural dataset contains 504 records representing 12 crop classes. The available observations provide numerical measurements for the six input features along with their associated crop labels.

The expected outcome is an interactive system in which a user enters the farm conditions and receives a recommended crop along with the model's prediction confidence.

The main assumptions are that the input measurements are representative of the field being evaluated and that the patterns present in the available dataset are sufficiently useful for learning crop-selection decisions.

The recommendation is treated as a data-driven decision-support result rather than a substitute for agricultural expertise.

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2. APPLICATION OF COURSE KNOWLEDGE

The solution applies concepts from classification, supervised learning, decision-tree modelling, model evaluation, and data visualization.

The six numerical attributes are used as predictor variables:

- → N – Nitrogen content
- → P – Phosphorus content
- → K – Potassium content
- → Rainfall – rainfall received
- → Temperature – ambient temperature
- → Humidity – atmospheric humidity

The crop attribute is used as the target variable.

Since the target consists of categorical crop names, a classification model is appropriate. A CART decision tree was selected because it can divide observations through a sequence of feature-based decisions and produce an interpretable classification structure.

The model uses Gini impurity to evaluate the quality of candidate splits. At each decision point, the tree selects feature thresholds that help separate observations belonging to different crop classes. Repeated splitting produces branches that ultimately lead to crop predictions.

The dataset was divided into 80% training data and 20% testing data using stratified sampling. Stratification was used to preserve the representation of the different crop classes in the two subsets.

Model performance was evaluated using classification accuracy, while feature importance and prediction probabilities were used to provide additional interpretation of the model.

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3. SOLUTION / DESIGN / METHODOLOGY

The developed system follows the workflow:
Dataset → Feature Selection → Train-Test Split → CART Model → Model Evaluation → User Input → Crop Prediction → Visualization

Step 1: Dataset Preparation
The agricultural dataset was loaded using Pandas. The six environmental and soil variables were selected as features, while the crop category was selected as the target.

Step 2: Train-Test Division
The available observations were divided into training and testing sets using an 80:20 ratio. A fixed random state was used to make the experiment reproducible, while stratification maintained the class distribution.

Step 3: CART Model Development
A DecisionTreeClassifier was used with:
Criterion = Gini Impurity
The model was trained using the six selected features.

Step 4: Model Evaluation
Predictions were generated for the testing set and compared with the actual crop labels. The resulting classification accuracy was used as the primary performance measure.

Step 5: Crop Recommendation
The interactive application accepts values for:
N, P, K, rainfall, temperature, and humidity.
These values are converted into the same feature structure used during training and supplied to the trained CART model. The predicted class is displayed as the recommended crop.

Step 6: Prediction Probability
The model also produces probabilities for the available crop classes. The five classes with the highest probabilities are displayed graphically to provide additional information about the prediction.

Step 7: Feature Importance
The importance values calculated by the trained decision tree are visualized to show the relative contribution of the input features to the model's decision process.

Step 8: CART Visualization
The decision tree is visualized using the major decision levels of the trained model. The visualization displays feature thresholds, Gini impurity, sample counts, and predicted classes, providing an interpretable representation of the classification process.

Step 9: Interactive Interface
A Streamlit-based interface was developed to combine the model, user inputs, prediction output, performance information, and visualizations into a single application.

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4. USE OF MODERN TOOLS

The implementation was developed using Python and the following tools:

Tool	Purpose
Python	Core implementation
Pandas	Dataset loading and data handling
Scikit-learn	CART classification, prediction, and accuracy evaluation
Matplotlib	Feature-importance and decision-tree visualization
Streamlit	Interactive crop recommendation interface

The application provides an interactive input panel where users can modify the six farm conditions. The recommendation is updated according to the supplied values.

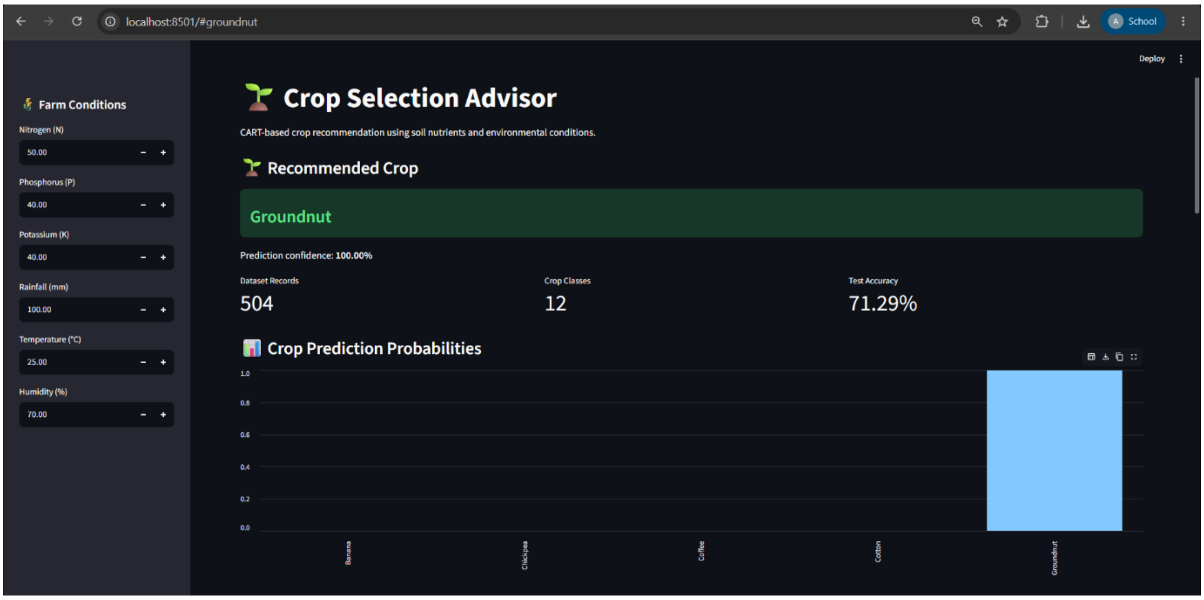
The implementation was executed successfully in a local Python environment. The interface also provides dataset information, prediction confidence, probability visualization, feature importance, and the CART decision-tree representation.

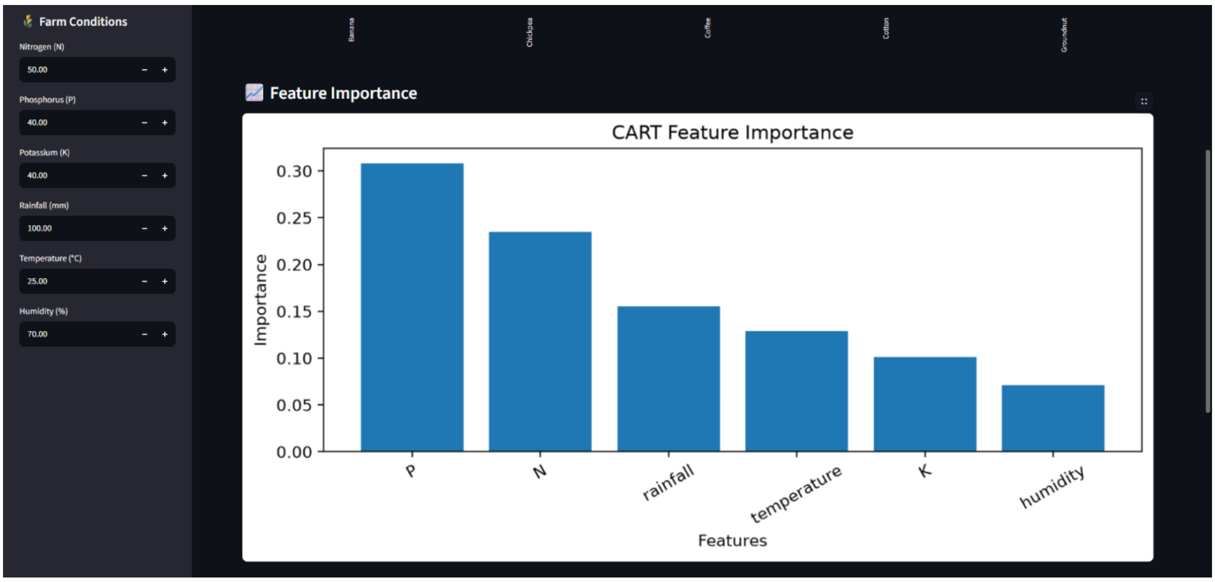
5. RESULTS AND VALIDATION

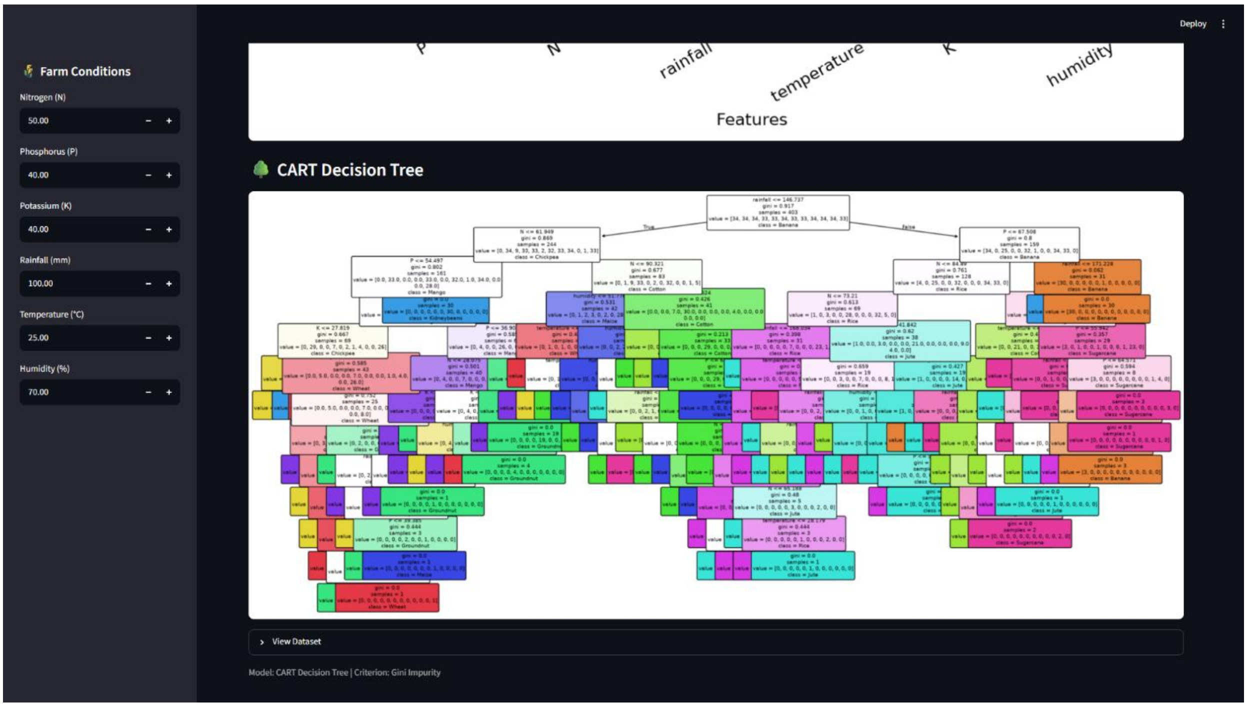
The developed Crop Selection Advisor successfully operated on the agricultural dataset and produced crop recommendations from user-supplied farm conditions.

Dataset and Model Results

Parameter	Obtained Result
Dataset records	504
Crop classes	12
Training proportion	80%
Testing proportion	20%
Test accuracy	71.29%
Classification criterion	Gini Impurity







The model achieved a test accuracy of 71.29%, indicating that the trained CART classifier was able to correctly classify a substantial proportion of the unseen testing observations.

Prediction Validation

The interactive application was tested with different combinations of soil and environmental conditions. The predicted crop changed when substantially different input conditions were supplied.

For example, the initial set of farm conditions produced a recommendation of Groundnut, while another set of randomly modified conditions produced Sugarcane. This confirms that the application is using the supplied feature values during prediction rather than displaying a fixed recommendation.

Prediction Probability

The probability visualization provides the relative likelihood assigned by the CART model to the available crop classes for a selected input. This provides more information than displaying only the final predicted class.

Feature Importance

The feature-importance visualization provides an indication of which of the six input variables contributed more strongly to the learned classification structure.

Decision Tree

The CART visualization demonstrates the model's hierarchical decision-making process. The displayed tree contains feature thresholds and Gini impurity values at its nodes, allowing the classification process to be interpreted rather than treated as a completely opaque prediction.

The tree display was restricted to its upper levels for readability while retaining the complete trained model for prediction.

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6. ANALYSIS AND ENGINEERING DECISION

The results demonstrate that CART can be used to construct a practical crop-classification system from soil and environmental measurements.

The 71.29% test accuracy indicates that the model learned meaningful relationships between the six input variables and the crop categories. However, the accuracy also indicates that the model does not classify every unseen observation correctly. Therefore, the recommendation should be interpreted as a model-based suggestion rather than a guaranteed crop outcome.

An important advantage of CART is its interpretability. Unlike a model whose decisions are difficult to inspect, a decision tree represents classification through a sequence of feature thresholds. The Gini impurity values shown at the nodes provide an indication of how the tree separates different crop classes.

The probability output provides an additional perspective on the prediction. Instead of relying only on the selected crop, the user can examine the relative probabilities of the leading alternatives.

The system also demonstrated sensitivity to the supplied conditions. Changing the farm inputs resulted in different crop recommendations, showing that the prediction is dependent on the combination of soil and environmental features.

The principal limitation is that the model's behaviour is constrained by the available dataset. The training data contains 504 observations and 12 crop classes, so predictions for conditions that differ substantially from the patterns represented in the dataset may be less reliable. The model also represents statistical patterns in the data and does not directly incorporate other agricultural factors such as soil type, crop economics, irrigation availability, pests, or farmer-specific requirements.

Considering these factors, CART was selected as an appropriate solution because it provides both classification capability and an interpretable decision structure while remaining practical for an interactive application.

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7. BROADER CONSIDERATIONS

The developed system has potential relevance to data-driven agricultural decision support.

From a sustainability perspective, considering soil and climatic conditions before selecting a crop can support more informed agricultural planning. Matching crops with suitable conditions may also help reduce inappropriate use of available agricultural resources.

From an environmental perspective, the system incorporates rainfall, temperature, and humidity along with soil nutrient information rather than relying solely on a crop-selection rule based on a single parameter.

From an economic perspective, an informed preliminary recommendation may help reduce the risk associated with selecting crops that are poorly suited to the observed conditions. However, economic decisions should also consider market prices, cultivation costs, expected yield, and local conditions.

From a societal and accessibility perspective, the Streamlit interface provides a straightforward mechanism for entering farm conditions and obtaining a recommendation without requiring the user to interact directly with the machine-learning code.

The system should be used responsibly because a machine-learning prediction depends on the quality and representativeness of the training data. Agricultural experts and local knowledge remain important when making actual cultivation decisions.

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8. CONCLUSION

A CART-based Crop Selection Advisor was successfully developed using soil nutrients and environmental conditions as predictors. The system uses N, P, K, rainfall, temperature, and humidity to classify the most suitable crop from the available crop categories.

The developed model was trained using an 80:20 train-test division and achieved a test accuracy of 71.29% on the available dataset. The implementation also provides crop prediction probabilities, feature importance, and an interpretable CART decision-tree visualization.

An interactive Streamlit interface was integrated with the model so that different farm conditions can be entered and corresponding crop recommendations can be generated dynamically. Testing confirmed that changing the input conditions can result in different crop recommendations.

The developed system demonstrates how fundamental data science and machine-learning techniques can be applied to an agricultural decision-support problem. Further improvement could involve increasing the diversity and size of the dataset, incorporating additional agricultural variables, tuning the decision-tree model, and comparing CART with other classification techniques.

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9. STUDENT REFLECTION

1. What did you learn from this assignment beyond what was directly taught in the classroom?
This assignment provided practical understanding of how a classification model can be transformed into a usable decision-support application. I learned that developing a machine-learning solution involves more than training a model; the data, evaluation method, visualization, user input, and interpretation of predictions are also important parts of the overall system.

2. What challenges did you face while developing the solution?
The main challenge was integrating the trained CART model with an interactive interface while ensuring that user-entered values were supplied to the model in the correct feature order. Another challenge was presenting the complete decision tree clearly, since an unrestricted tree becomes highly cluttered and difficult to interpret.

3. How did visualization improve your understanding of the model?
The feature-importance graph helped identify the relative contribution of the input variables, while the decision-tree visualization showed how the model divides observations using feature thresholds and Gini impurity. The probability graph also provided a better understanding of the alternatives considered by the classifier.

4. What did you learn from testing different farm conditions?
Testing different combinations demonstrated that the recommendation depends on the supplied environmental and soil conditions. Changing the input values can move an observation into a different region of the learned decision tree and consequently produce a different crop recommendation.

5. What would you improve if additional time and resources were available?
I would evaluate the model using additional performance measures and compare CART with other suitable classification algorithms. I would also use a larger and more diverse agricultural dataset to improve the reliability of recommendations across different farming conditions.

6. How could the system be made more useful for real-world agricultural decision-making?
The system could be enhanced by incorporating additional information such as soil type, geographical location, irrigation availability, seasonal conditions, crop prices, and historical yield. Combining these factors with expert agricultural knowledge could make the recommendation more comprehensive.

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10. REFERENCES

1. Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and Regression Trees*. Wadsworth International Group.

2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

3. McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*, 51–56.

4. Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering, 9*(3), 90–95.

5. Streamlit Documentation. *Streamlit: A Faster Way to Build and Share Data Apps*. Accessed during implementation.

6. Agricultural crop dataset (crop_data.csv) used for training and testing the Crop Selection Advisor. The dataset contains N, P, K, rainfall, temperature, humidity, and crop attributes.

7. AI-assisted tools were used as supporting resources during the development and documentation process.

COMMON ASSESSMENT RUBRIC

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

CO-PO MAPPING

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

FACULTY EVALUATION & CONTINUOUS IMPROVEMENT

CO ATTAINMENT

Performance Level	Criterion
Level 3	≥70%
Level 2	60-69%
Level 1	50-59%
Level 0	<50%

Target CO Attainment:
Actual CO Attainment:

Faculty Analysis:

Areas in which students performed well: Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:
